



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Preeta Sundara Shetty

16-July-2026

<https://github.com/Preeta-DE/SpaceX-IBM-Capstone>



Outline

- Executive Summary
- Introduction
- Methodology: Data Collection, Data Wrangling, EDA Analytics, Predictive Analytics
- Results: EDA with Visualization, EDA with SQL, Folium, Plotly Dash, and Predictive Analytics
- Conclusion
- Appendix

Executive Summary

- Built an end-to-end SpaceX Falcon 9 landing prediction pipeline using API data, web scraping, data wrangling, SQL, visualization, interactive analytics, and machine learning.
- Created a binary landing outcome label where successful landings were encoded as 1 and unsuccessful outcomes as 0.
- EDA showed that launch site, orbit type, payload mass, and operational maturity influence landing success patterns.
- SQL analysis highlighted four unique launch sites, 45,596 kg NASA CRS payload mass, and the first successful ground landing on 2015-12-22.
- Interactive analytics were delivered through Folium maps and a Plotly Dash dashboard with site filters, pie charts, and payload-based scatter plots.
- Classification models reached 83.33% test accuracy, with Decision Tree achieving the highest cross-validation score during tuning.

Introduction

SpaceX advertises Falcon 9 launches at about \$62M, while competitor launches can exceed \$165M; first-stage reusability is a major reason for this cost advantage. By determining if the first stage will land, we can determine the price of the launch. To do this, we can use public data and machine learning models to predict whether SpaceX, or a competing company, can reuse the first stage.

Business question: can we predict whether the Falcon 9 first stage will land successfully using historical launch data?

Project objective: collect launch data, prepare it for analysis, discover patterns, build interactive analytics, and train classification models for landing prediction.

Main analytical questions:

Which sites and orbit types have higher success rates?

How do payload and flight history relate to outcomes?

Which model best predicts landing success?

Section 1

Methodology

Methodology

- Collected historical launch data from the SpaceX REST API and complementary records from Wikipedia using web scraping.
- Wrangled the dataset by cleaning fields, standardizing outcomes, and deriving the target class for supervised learning.
- Performed EDA using seaborn/matplotlib visualizations and SQLite SQL queries to identify operational patterns.
- Built interactive analytics using Folium maps and a Plotly Dash dashboard for site and payload exploration.
- Prepared features, standardized data, tuned multiple classification models with GridSearchCV, and compared model performance

Data Collection

- Two complementary collection methods were used: SpaceX API calls for structured launch metadata and Wikipedia scraping for historical launch records.
- The API workflow extracted rocket, payload, core, launchpad, latitude, longitude, orbit, and landing-related fields.
- The scraping workflow parsed the Falcon 9 / Falcon Heavy launch history table into a pandas dataframe for downstream analysis.
- GitHub reference: <https://github.com/Preeta-DE/SpaceX-IBM-Capstone>

Data Collection – SpaceX API

- Requested past launches from the SpaceX REST API endpoint and normalized the JSON response into a dataframe.
- Used helper functions to retrieve booster version, payload mass, orbit, launch site, coordinates, landing outcome, flights, gridfins, reuse status, legs, landing pad, block, reuse count, and serial.
- This API-driven dataset provided the structured foundation for wrangling, EDA, geospatial analysis, and machine learning.
- Notebook reference: https://github.com/Preeta-DE/SpaceX-IBM-Capstone/blob/main/01_SpaceX-Data-collection-api.ipynb

API workflow

1. Call SpaceX past launches endpoint
2. Normalize JSON into pandas dataframe
3. Filter single-core / single-payload Falcon 9 launches.
4. Enrich with rocket, payload, core, and launchpad details after replacing the missing values with mean().
5. Build analysis-ready launch dataset and export to csv file.

Data Collection - Scraping

- Used requests and BeautifulSoup to scrape Falcon 9 and Falcon Heavy launch records from a static Wikipedia snapshot.
- Parsed launch date, booster version, launch site, payload, payload mass, orbit, customer, launch outcome, and booster landing fields.
- Converted the HTML table into a pandas dataframe and exported it for downstream use.
- Notebook reference:
https://github.com/Preeta-DE/SpaceX-IBM-Capstone/blob/main/O2_SpaceX-Web scraping.ipynb

Web scraping workflow

1. Request static Wikipedia launch-history page
2. Parse Falcon 9 / Falcon Heavy HTML table
3. Extract mission, payload, orbit, customer, and outcome fields
4. Clean text columns and standardize values
5. Export structured dataframe to csv file for downstream analysis

Data Wrangling

- Cleaned launch records and inspected launch site counts, orbit counts, and landing outcomes.
- Defined unsuccessful outcomes as {False ASDS, False Ocean, False RTLS, None ASDS, None None}.
- Created the classification target column Class, where 1 = successful landing and 0 = unsuccessful landing.
- Prepared the cleaned dataset for EDA, SQL analysis, dashboarding, and predictive modeling.
- Notebook reference: https://github.com/Preeta-DE/SpaceX-IBM-Capstone/blob/main/O3_SpaceX-Data-wrangling.ipynb

EDA with Data Visualization

- Used seaborn and matplotlib to explore relationships between launch outcome and operational variables.
- Plotted Flight Number vs Launch Site, Payload vs Launch Site, Success rate by Orbit, Flight Number vs Orbit, Payload vs Orbit, and yearly success trend.
- These visualizations helped identify patterns in experience, site behavior, orbit performance, and payload-related success.
- Notebook reference: https://github.com/Preeta-DE/SpaceX-IBM-Capstone/blob/main/O5_EDA_Data_Visualization.ipynb

EDA with SQL

- Loaded launch data into SQLite to answer business-oriented analytical questions using SQL.
- Queried unique launch sites, NASA payload totals, average payload by booster version, first successful ground landing date, mission outcomes, and other filtered launch records.
- SQL results were used to summarize operational evidence in a concise, presentation-friendly format.
- Notebook reference: https://github.com/Preeta-DE/SpaceX-IBM-Capstone/blob/main/O4_EDA_SQL_sqlite.ipynb

Build an Interactive Map with Folium

- Marked and labeled all launch sites on a map using latitude and longitude coordinates.
- Added color-labeled success and failure markers for each launch and explored proximity to infrastructure such as coastline, railway, and highway.
- This geospatial analysis supported the insight that launch-site location and nearby features matter operationally.
- Notebook reference: https://github.com/Preeta-DE/SpaceX-IBM-Capstone/blob/main/06_SpaceX_VisualAnalytics_Folium.ipynb

Build a Dashboard with Plotly Dash

- Developed an interactive dashboard with a launch-site dropdown, a success pie chart, a payload range slider, and a payload-success scatter plot.
- Users can compare all sites or isolate a single site to explore success patterns by payload and booster category.
- The dashboard complements static EDA by enabling interactive filtering and visual exploration.
- Python file reference: https://github.com/Preeta-DE/SpaceX-IBM-Capstone/blob/main/07_SpaceX_Visual_Analytics_Plotly_Dash.py

Predictive Analysis (Classification)

- Prepared features using one-hot encoding and standardized numeric inputs before model training.
- Split data into training and test sets, then used GridSearchCV to tune Logistic Regression, SVM, Decision Tree, and KNN models.
- Evaluated models with accuracy and confusion matrices to determine the strongest landing predictor.
- Identified the best model using Jaccard_Score, F1_Score and Accuracy.
- Notebook reference: https://github.com/Preeta-DE/SpaceX-IBM-Capstone/blob/main/O8_SpaceX_PredictiveAnalytics.ipynb

Results

Exploratory Data Analysis:

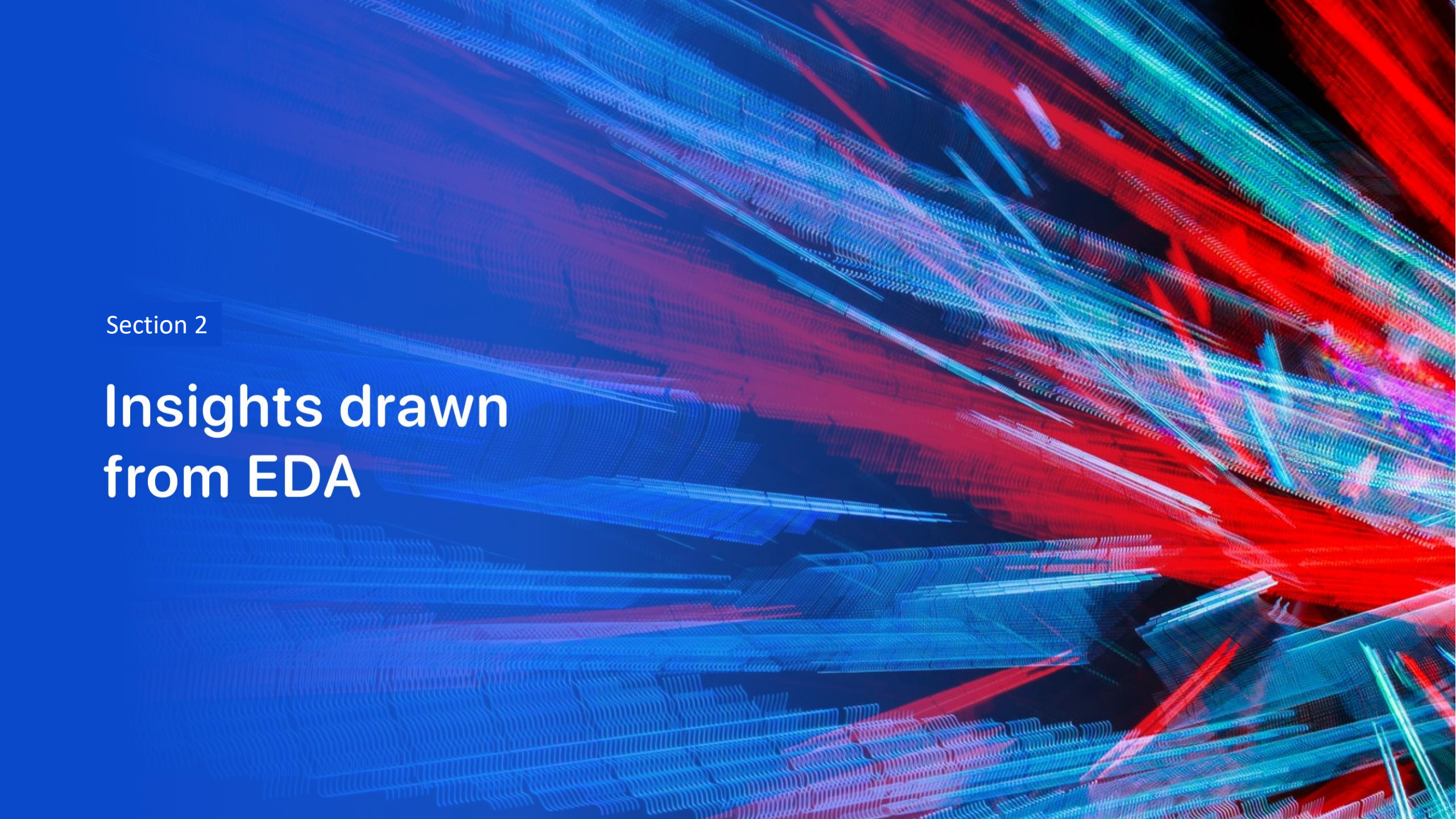
- Launch success has improved over time
- KSC LC-39A has the highest success rate among landing sites
- Orbits ES-L1, GEO, HEO and SSO have a 100% success rate

Visual Analytics:

- Most launch sites are near the equator, and all are close to the coastline.
- Launch sites are far away enough from anything a failed launch can damage (city, highway, railway), while still close enough to bring people and material to support launch activities.

Predictive Analytics:

- Decision Tree model is the best predictive model for the dataset

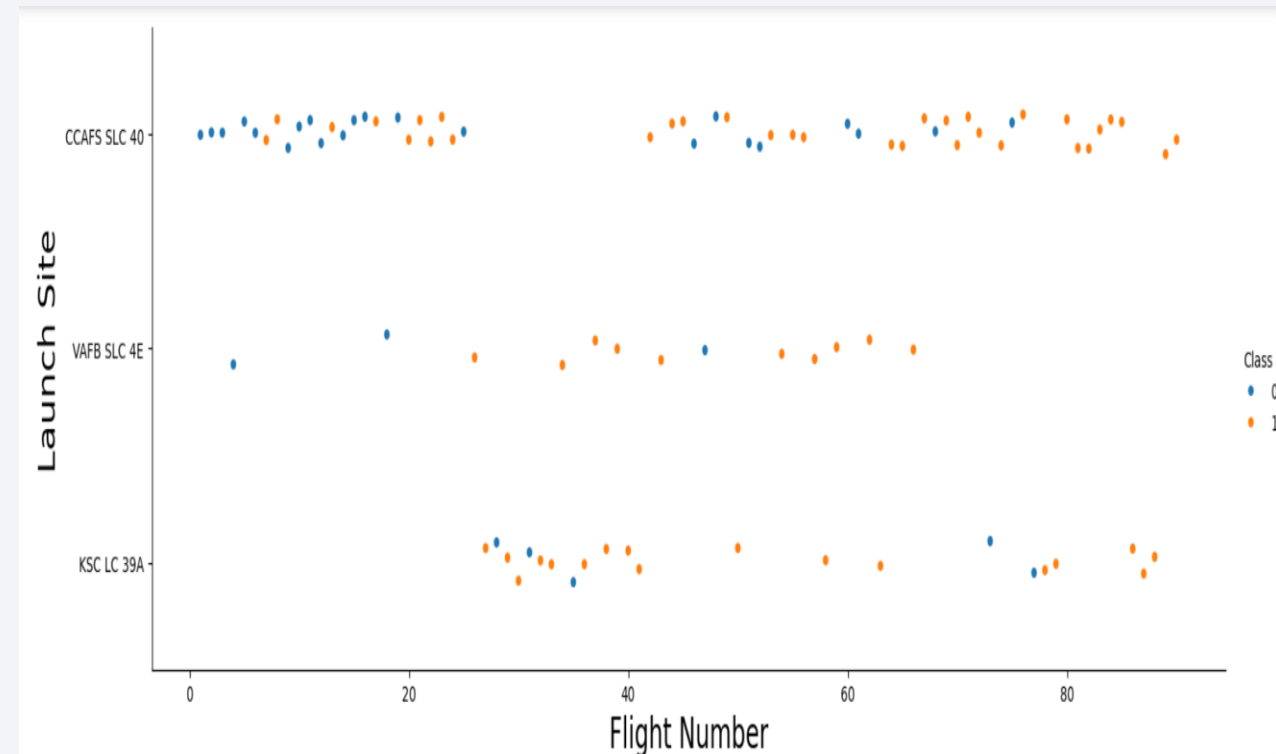
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is high-tech and digital.

Section 2

Insights drawn from EDA

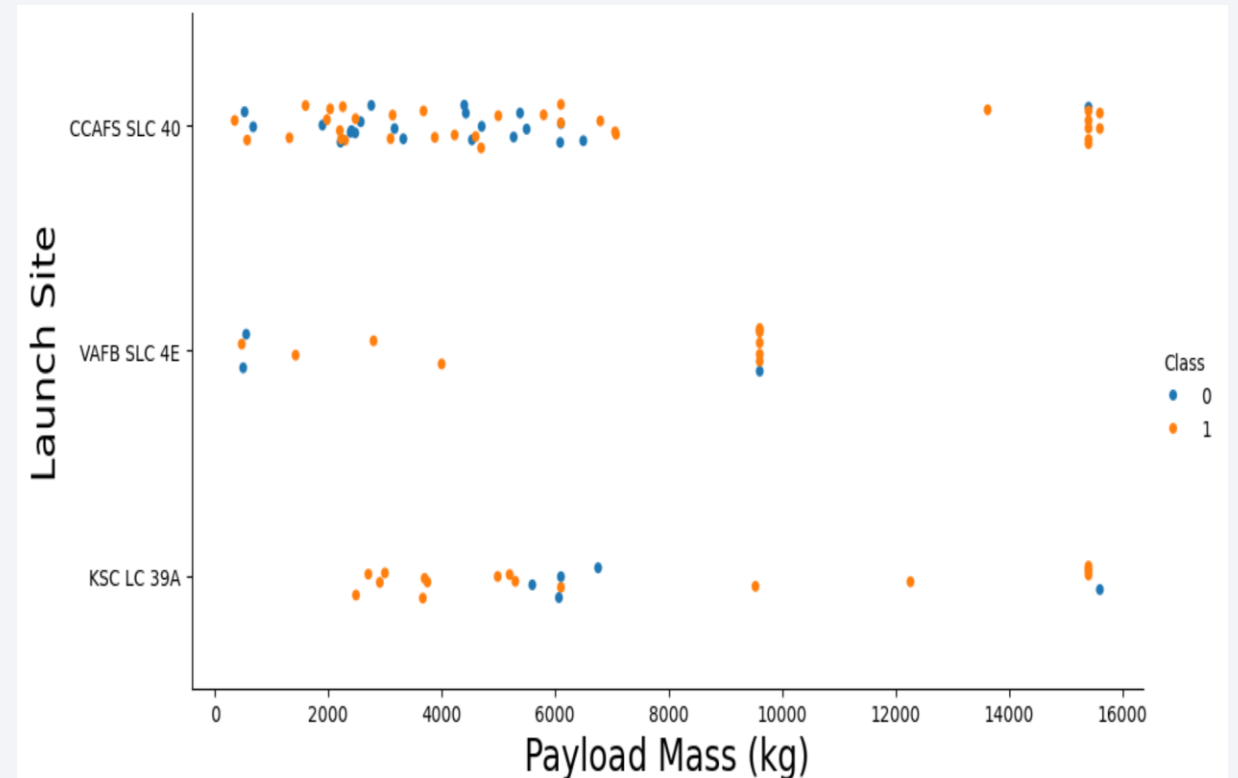
Flight Number vs. Launch Site

- Earlier flights had a lower success rate (blue = fail)
- Later flights had a higher success rate (orange = success)
- Around half of launches were from CCAFS SLC 40 launch site
- VAFB SLC-4E and KSC LC-39A have higher success rates
- We can infer that new launches have a higher success rate
- Insight : Mission experience improved first-stage landing reliability



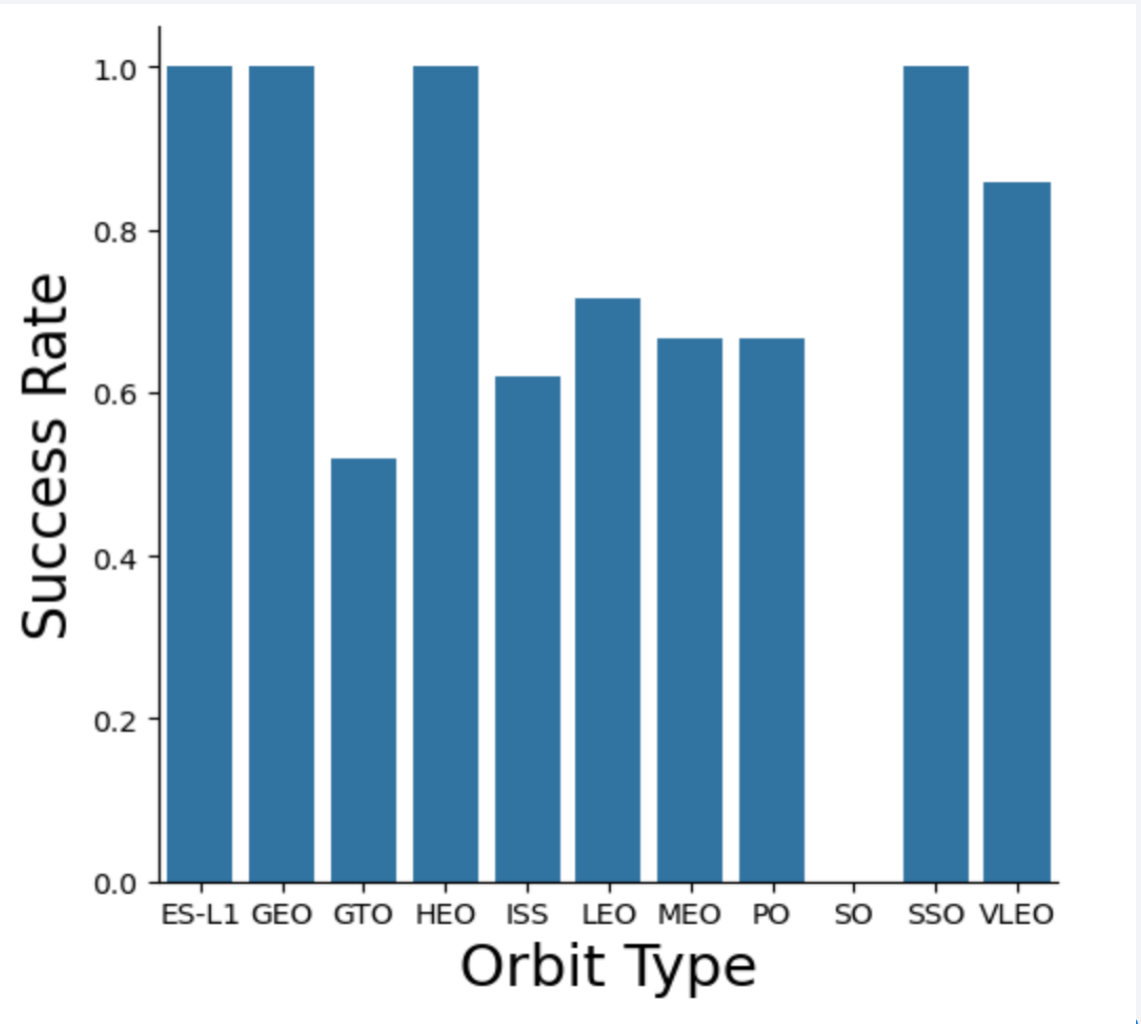
Payload vs. Launch Site

- Typically, the higher the payload mass (kg), the higher the success rate
- Most launches with a payload greater than 7,000 kg were successful
- KSC LC-39A has a 100% success rate for launches less than 5,500 kg
- VAFB SLC-4E has not launched anything greater than ~10,000 kg
- Insight: Higher payload mass has higher success rate



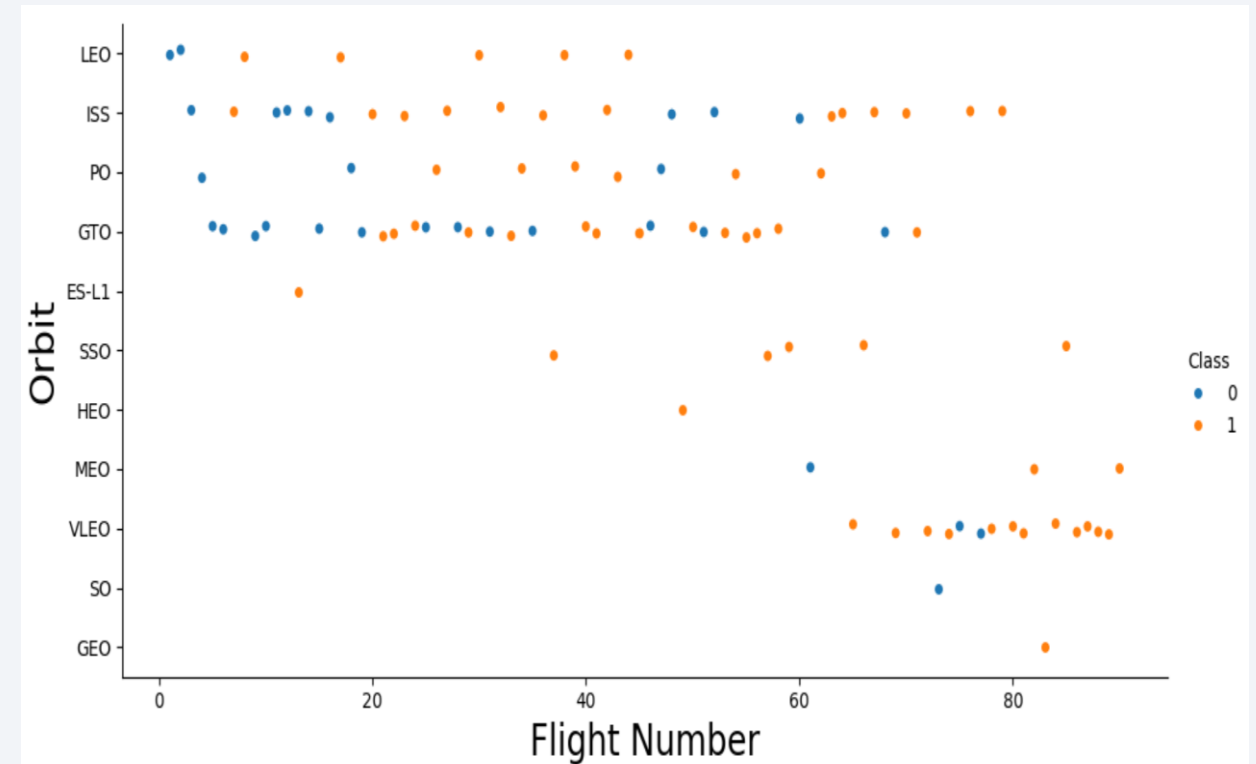
Success Rate vs. Orbit Type

- ES-L1, GEO, HEO and SSO orbit categories show consistently stronger success rates than others, indicating mission profile matters.
- Insight: Orbit should be treated as an important predictive feature.



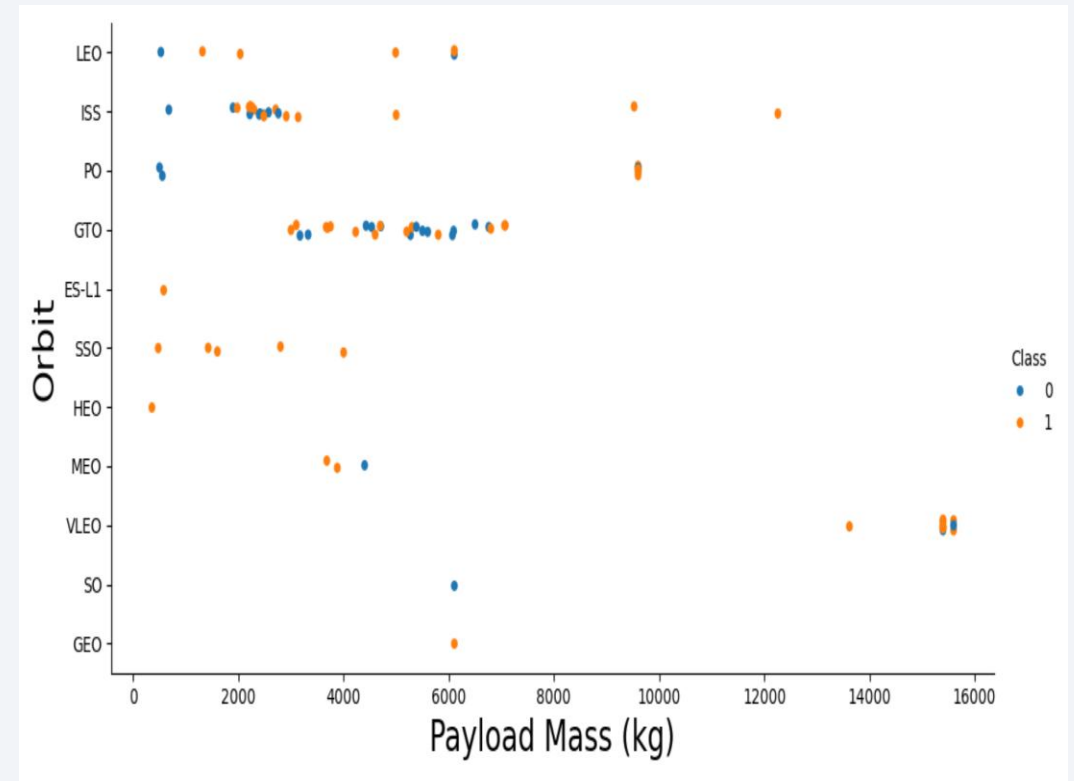
Flight Number vs. Orbit Type

- Higher flight numbers generally coincide with improved outcomes across several orbit types.
- This suggests that maturation of operations helped SpaceX execute more demanding mission profiles successfully.
- Insight: Time and experience interact with orbit complexity.



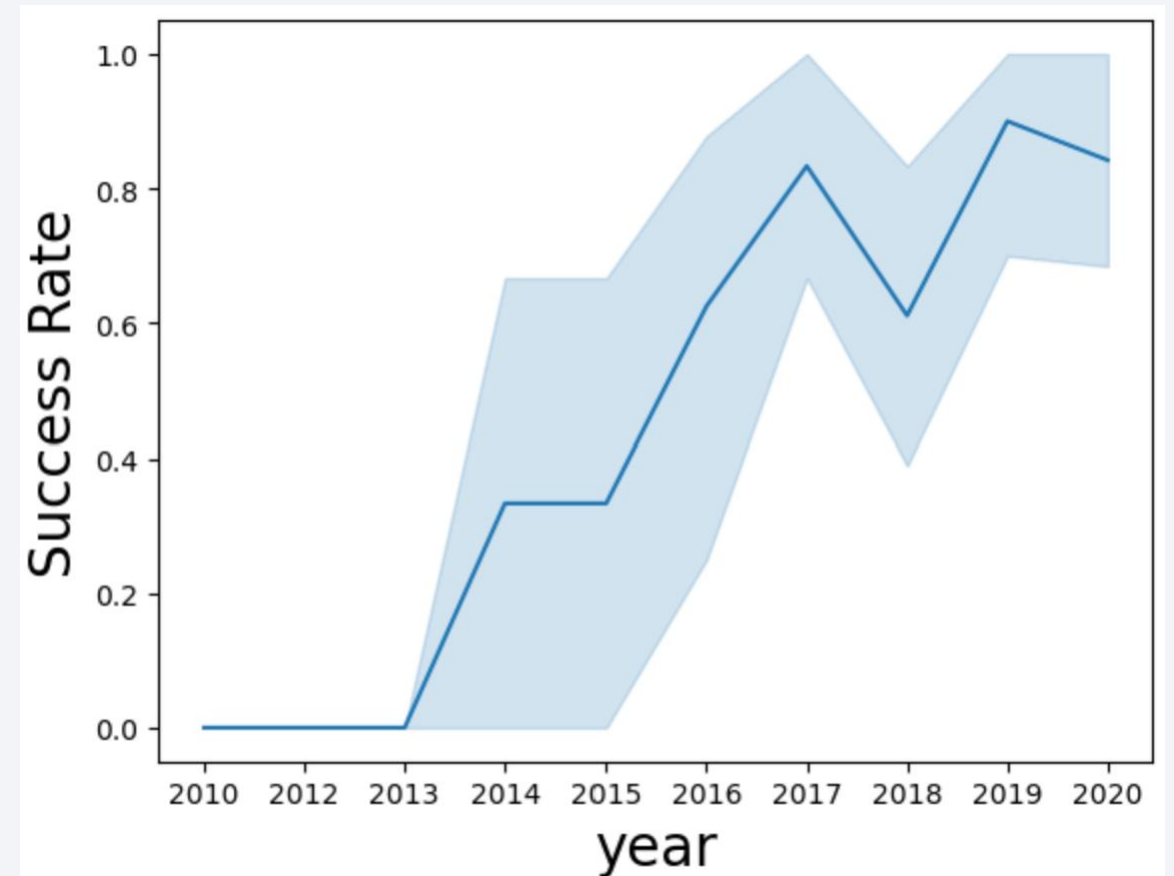
Payload vs. Orbit Type

- Heavy payloads are better with LEO, ISS and PO orbits.
- The GTO orbit has mixed success with heavier payloads.
- Insight: Payload and Orbit jointly influence the likelihood of successful recovery.



Launch Success Yearly Trend

- The yearly average success rate trends upward over time.
- This supports the interpretation that SpaceX improved landing capability through reuse, design iteration, and operational experience.
- Insight: Later missions have higher landing success than early ones.



All Launch Site Names

- Unique launch sites identified in the SQL analysis: CCAFS LC-40, CCAFS SLC-40, VAFB SLC-4E, and KSC LC-39A.

Display the names of the unique launch sites in the space mission

```
%sql SELECT DISTINCT(Launch_Site) FROM SPACEXTABLE
```

```
* sqlite:///my_data1.db
```

Done.

Launch_Site
CCAFS LC-40
CCAFS SLC-40
VAFB SLC-4E
KSC LC-39A

Launch Site Names Begin with 'CCA'

- Displays 5 records of Launch Site Names beginning with CCA.

Display 5 records where launch sites begin with the string 'CCA'

```
%sql SELECT * FROM SPACEXTABLE WHERE Launch_Site like "%CCA%" LIMIT 5
```

```
* sqlite:///my_data1.db
```

Done.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- Total payload carried by boosters for NASA CRS missions: 45,596 kg.

Display the total payload mass carried by boosters launched by NASA (CRS) ⓘ

```
%sql SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE CUSTOMER = 'NASA (CRS)'
```

```
* sqlite:///my_data1.db
```

Done.

SUM(PAYLOAD_MASS__KG_)

45596

Average Payload Mass by F9 v1.1

- Average of 2928.4 kg was carried by booster version F9 v1.1

Display average payload mass carried by booster version F9 v1.1

```
%sql Select AVG(PAYLOAD_MASS__KG_) from SPACEXTABLE WHERE Booster_Version = 'F9 v1.1'
```

```
* sqlite:///my_data1.db
```

Done.

AVG(PAYLOAD_MASS__KG_)

2928.4

First Successful Ground Landing Date

- The first successful landing outcome on a ground pad occurred on 2015-12-22.

```
%sql SELECT min(DATE) FROM SPACEXTABLE WHERE Landing_Outcome like 'Success%ground pad%'
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
min(DATE)
```

```
2015-12-22
```

Successful Drone Ship Landing with Payload between 4000 and 6000

- Boosters listed by SQL for successful drone ship landings with payload mass between 4,000 and 6,000 kg included F9 FT B1022, F9 FT B1026, F9 FT B1021.2, and F9 FT B1031.2.
- These records show that successful drone ship recovery was achieved even for relatively heavy payload missions.

```
List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000 ⓘ

T Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome like 'Success (drone ship)%' and PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000

* sqlite:///my_data1.db
Done.

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2
```

Total Number of Successful and Failure Mission Outcomes

- SQL mission outcome counts show Success as the dominant mission outcome (98), with only a small number of failures or partial failures.
- This confirms the high overall mission reliability of the Falcon program in the analyzed records.

List the total number of successful and failure mission outcomes

```
%sql SELECT Mission_Outcome, Count(Mission_Outcome) FROM SPACEXTABLE GROUP BY Mission_Outcome
```

```
* sqlite:///my_data1.db
```

Done.

Mission_Outcome	Count(Mission_Outcome)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- SQL identified several Block 5 boosters among the boosters carrying the maximum payloads.
- Examples include B1048.4, B1049.4, B1051.3, and later reuse variants.

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
%sql SELECT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS_KG_ IN (SELECT MAX(PAYLOAD_MASS_KG_) FROM SPACEXTABLE)
```

```
* sqlite:///my_data1.db
```

Done.

Booster_Version

F9 B5 B1048.4

F9 B5 B1049.4

F9 B5 B1051.3

F9 B5 B1056.4

F9 B5 B1048.5

F9 B5 B1051.4

F9 B5 B1049.5

F9 B5 B1060.2

F9 B5 B1058.3

F9 B5 B1051.6

F9 B5 B1060.3

F9 B5 B1049.7

2015 Launch Records

- In 2015, SQL identified failed drone-ship landing outcomes including boosters F9 v1.1 B1012 and F9 v1.1 B1015 from CCAFS LC-40

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
ome, Booster_Version, Launch_Site from SPACEXTABLE WHERE Landing_Outcome like "Failure (drone ship)%" and substr(Date,0,5)="2015"
```

```
* sqlite:///my_data1.db
```

Done.

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Between the date 2010-06-04 and 2017-03-20, the Failure (drone ship) count of Landing Outcome is 5 whereas Success (ground pad) is 3.

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
Landing_Outcome like "Success (ground pad)" and Date between "2010-06-04" and "2017-03-20" group by Landing_Outcome order by Count(*) desc
```

* sqlite:///my_data1.db

Done.

Landing_Outcome	Landing_Outcome_Count
Failure (drone ship)	5
Success (ground pad)	3

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

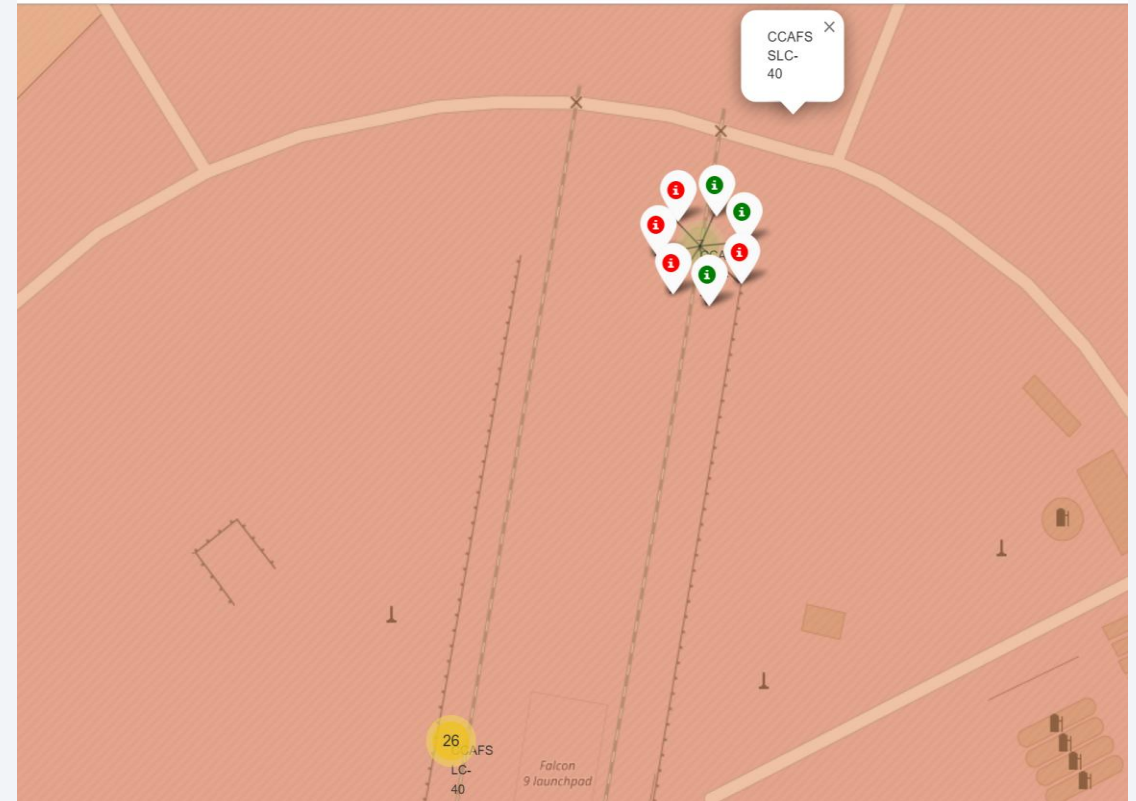
Launch Sites with Folium Markers

- All major launch sites are positioned near the coast.
- This reduces public safety risk, since rockets carry large amounts of fuel and any failure is less likely to affect cities or residential areas.



Launch Outcomes

- At each launch site, green markers indicate successful launches and red markers indicate unsuccessful launches.
- At CCAFS SLC-40, the displayed sample highlights a 3/7 success rate, or 42.9%.



Distance to Proximities

CCAFS SLC-40 launch site:

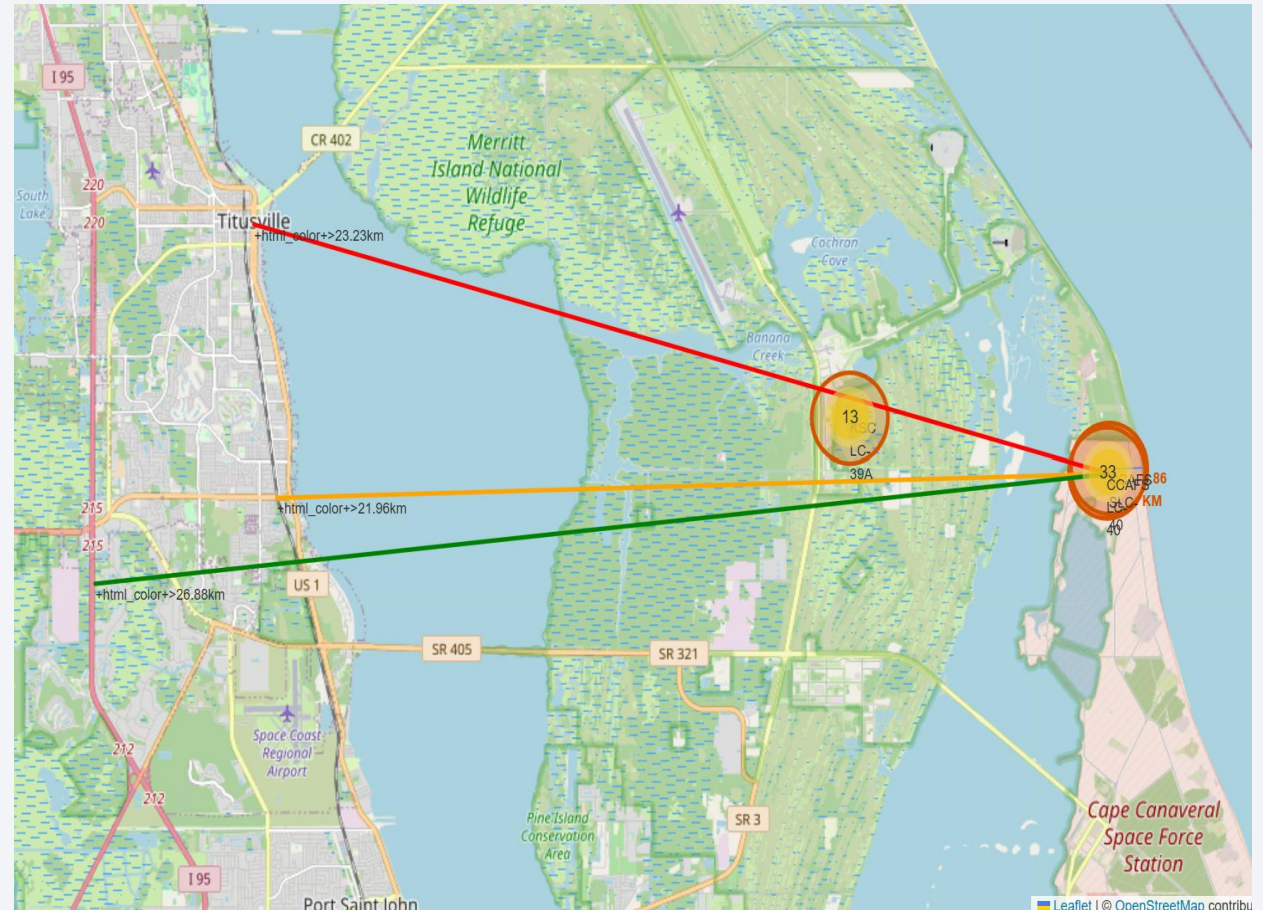
0.86 km distance from the nearest coastline.

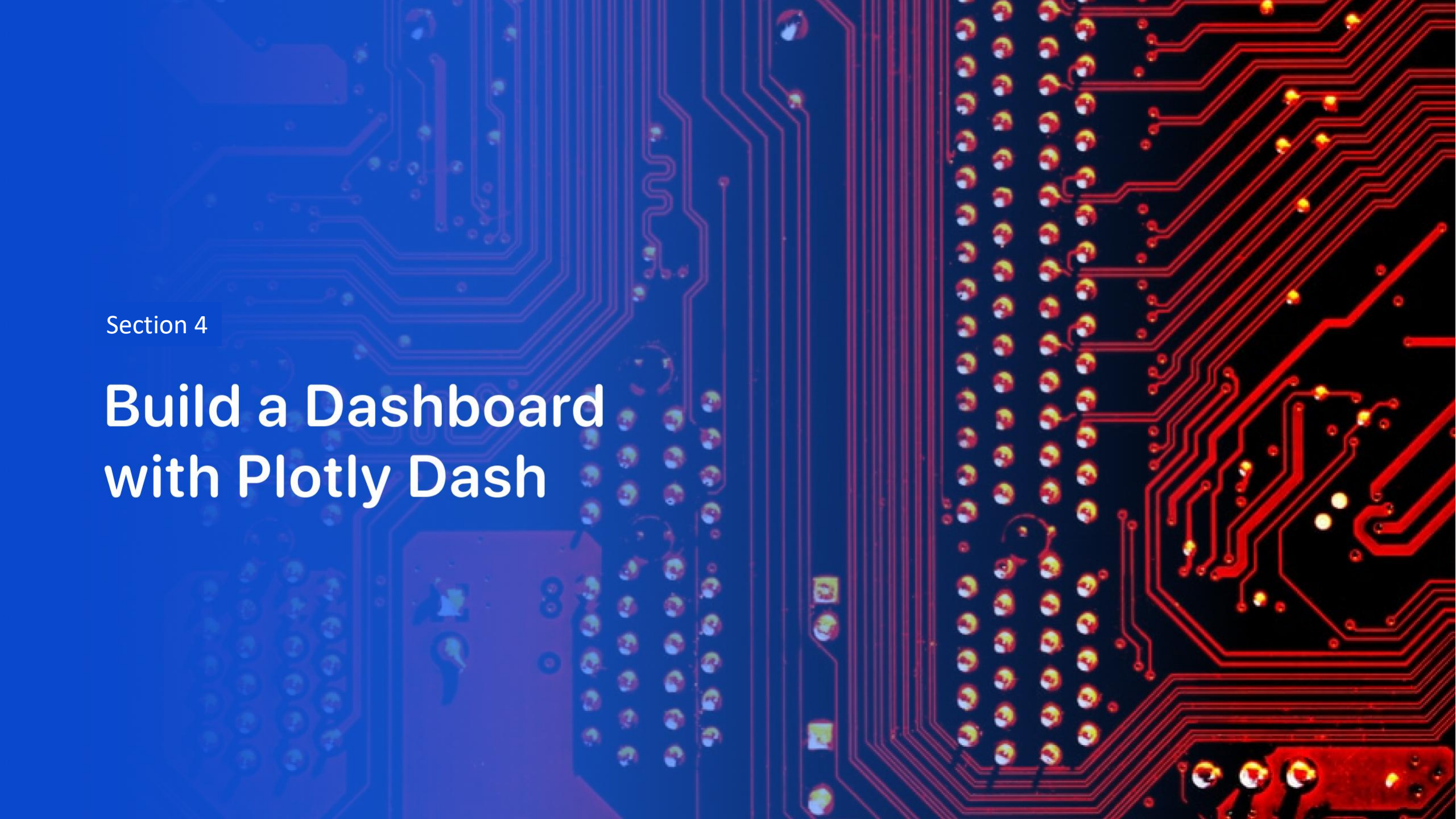
21.96 km distance from the nearest railway.

23.23 km distance from the nearest city.

26.88 km distance from the nearest highway.

These features matter because launch sites require both safety buffers and logistical accessibility



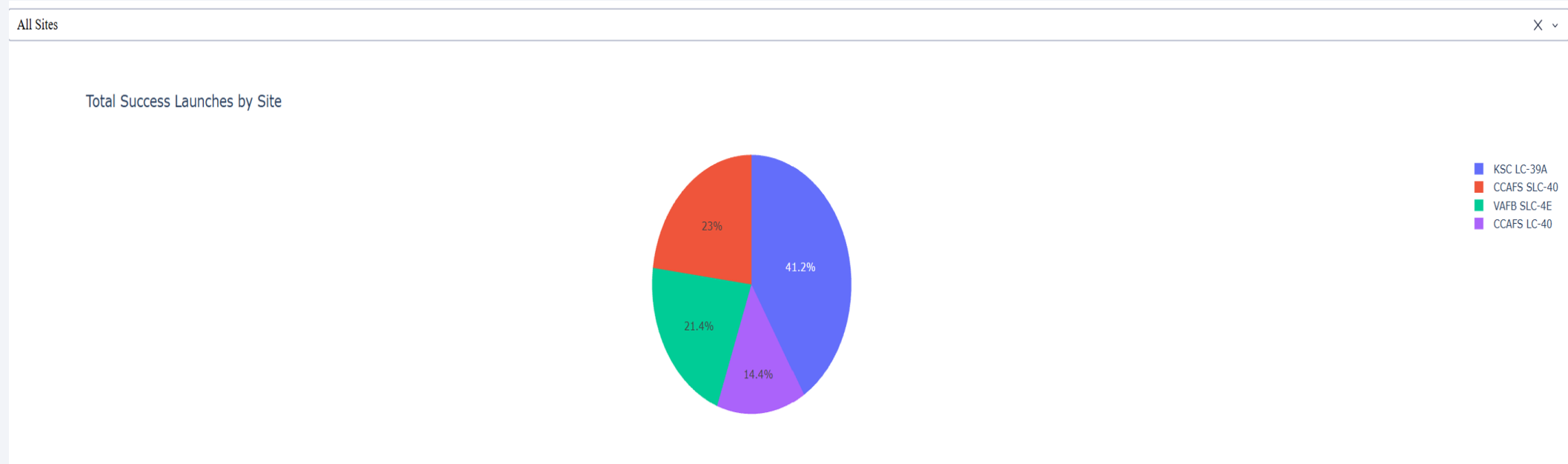


Section 4

Build a Dashboard with Plotly Dash

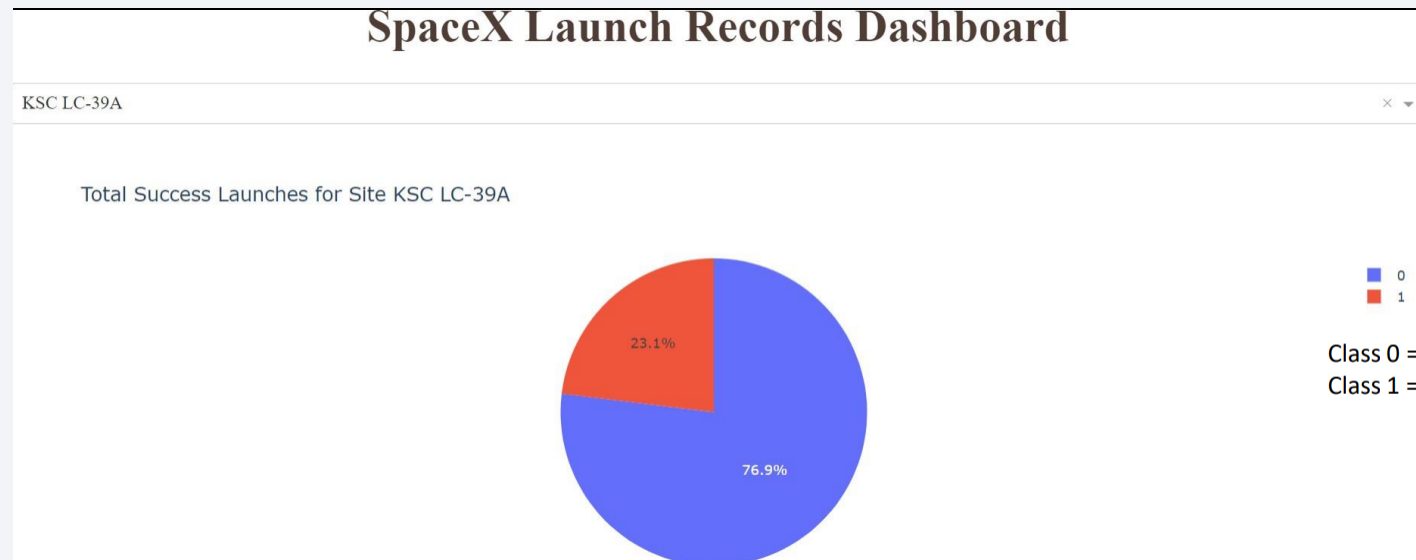
Launch Success by Site

- The Plotly Dash pie chart summarizes launch success across all sites using a site-selection dropdown.
- KSC LC-39A has the most successful launches amongst launch sites (41.2%)



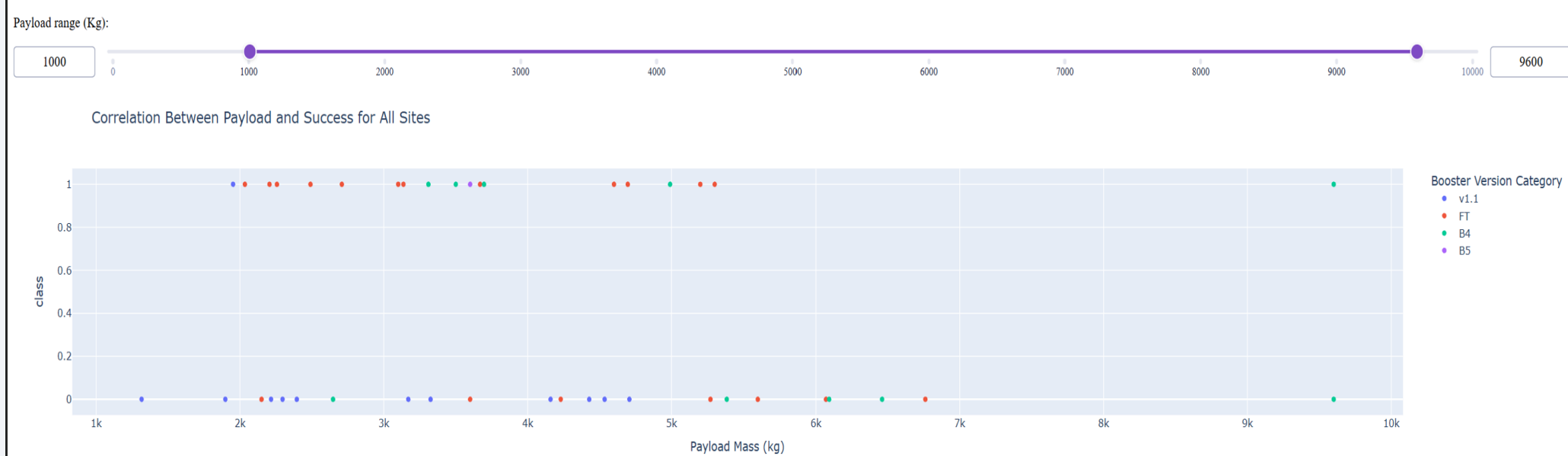
Launch Success (KSC LC-39A)

- KSC LC-39A has the highest success rate amongst launch sites - 76.9%
- 10 successful launches and 3 failed launches.



Payload Mass and Success

- Payloads between 2,000 kg and 5,000 kg have the highest success rate.
- Class 1 indicates success and 0 indicates failure.



Section 5

Predictive Analysis (Classification)

Classification Accuracy

- Test accuracy for Logistic Regression, SVM, Decision Tree, and KNN was 83.33% in the notebook outputs.
- During tuning, the Decision Tree reached the highest cross-validation score at 0.875, while Logistic Regression, SVM, and KNN were close behind.
- Decision Tree model slightly outperformed the rest

Model selection note: all four models tied on test accuracy, but Decision Tree was chosen because it achieved the best cross-validation score during tuning.

	LogReg	SVM	Tree	KNN
Jaccard_Score	0.800000	0.800000	0.800000	0.800000
F1_Score	0.888889	0.888889	0.888889	0.888889
Accuracy	0.833333	0.833333	0.833333	0.833333

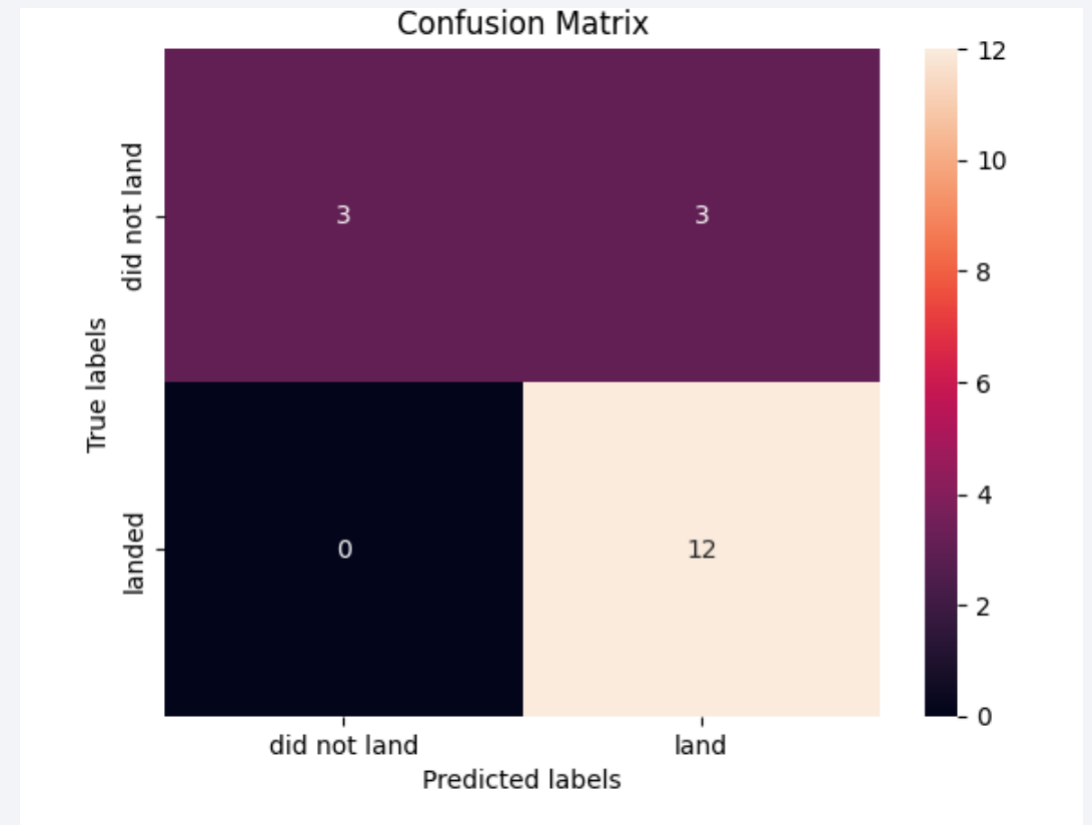
```
models = {'KNeighbors': knn_cv.best_score_,
          'DecisionTree': tree_cv.best_score_,
          'LogisticRegression': logreg_cv.best_score_,
          'SupportVector': svm_cv.best_score_}

bestalgorithm = max(models, key=models.get)
print('Best model is', bestalgorithm, 'with a score of', models[bestalgorithm])
if bestalgorithm == 'DecisionTree':
    print('Best params is :', tree_cv.best_params_)
if bestalgorithm == 'KNeighbors':
    print('Best params is :', knn_cv.best_params_)
if bestalgorithm == 'LogisticRegression':
    print('Best params is :', logreg_cv.best_params_)
if bestalgorithm == 'SupportVector':
    print('Best params is :', svm_cv.best_params_)

Best model is DecisionTree with a score of 0.875
Best params is : {'criterion': 'entropy', 'max_depth': 2, 'max_features': 'sqrt', 'min_samples_leaf': 1,
```

Confusion Matrix

- The confusion matrix shows how well the classifier predicted landing outcomes on the test set.
- Correct predictions: 12 successful landings and 3 failed landings.
- Errors: 3 failed launches were incorrectly predicted as successful landings; no actual successful landings were missed.
- Overall accuracy: 83.3%



Conclusions

Key takeaways

- Built a complete landing-prediction workflow from API collection and scraping through wrangling, EDA, SQL, mapping, dashboarding, and classification.
- Landing reliability improved over time, showing that operational maturity is strongly linked to successful recovery.
- KSC LC-39A showed the strongest site-level landing performance, while orbit type and payload characteristics also influenced success.
- Folium proximity analysis showed why launch sites balance safety and logistics: coastal placement reduces risk while preserving access.
- All four classifiers reached 83.33% test accuracy; Decision Tree was selected as the best tuned model based on cross-validation.
- Business implication: stronger landing prediction improves launch-cost estimation and helps explain SpaceX's reusable-launch advantage.

Appendix

- Supporting assets included in the GitHub repository and project deliverables:
- <https://github.com/Preeta-DE/SpaceX-IBM-Capstone>

Thank you!

